

GPU A+ Server AS -4125GS-TNRT2

DP AMD 4U Dual-Root PCIe GPU System with up to 10 PCIe GPUs



Key Applications

AI / Deep Learning, Big Data Analytics and AI, High Performance Computing, Engineering/scientific research,

Key Features

- Flexible GPU support: active & passive GPUs
Dual Root Configuration with PCIe Switch for up to 10 double width, full length GPUs
8 Hot-Swap Heavy Duty Fans with Optimal Fan Speed Control;
- Dual-Socket, AMD EPYC™ 9004/9005 Series Processors up to 400W TDP;
- Supports up to 24 DIMM Slots, 6000 MT/s 6TB DDR5 memory (Please check System Memory Section for detail);
- AIOM / OCP 3.0 Support;
- Total of 8 Hot-swap 2.5" NVMe drive bays, 1 onboard Gen3 M.2 Slot;
- Total of 4x 2000W Redundant (2+2 or 3+1) Titanium Level power supplies (2+2 under 4000 Watt, 3+1 under 6000 Watt)
(Power supply full redundancy based on configuration and application load);



Form Factor	4U Rackmount Enclosure: 437 x 178 x 737mm (17.2" x 7" x 29") Package: 685 x 675 x 1041mm (27" x 26.57" x 41")
Processor	Dual processor(s) AMD EPYC™ 9004/9005 Series Processors (* AMD EPYC™ 9005 Series drop-in support requires board revision 2.x) Up to 160C/320T; Up to 512MB Cache per CPU
GPU	Max GPU Count: Up to 10 double-width GPUs Supported GPU: NVIDIA PCIe: L4, L40S, NVIDIA RTX PRO™ 4500 Blackwell Server Edition, NVIDIA RTX PRO™ 5000 Blackwell, NVIDIA RTX PRO™ 6000 Blackwell Max-Q Workstation Edition, RTX 4000 Ada Generation, RTX 4000 SFF Ada Generation, RTX 4500 Ada, RTX 5000 Ada, RTX 6000 Ada Generation AMD PCIe: Instinct™ MI210 CPU-GPU Interconnect: PCIe 5.0 x16 Switch Dual-Root GPU-GPU Interconnect: NVIDIA® NVLink™ Bridge (optional), AMD Infinity Fabric™ Link (optional)
System Memory	Slot Count: 24 DIMM slots Max Memory (1DPC): Up to 6TB 4800MT/s ECC DDR5 RDIMM (AMD EPYC™ 9004 Series Processor) Max Memory (1DPC): Up to 6TB 6000MT/s ECC DDR5 RDIMM (AMD EPYC™ 9005 Series Processor)
Drive Bays Configuration	Default: Total 10 bays 8 front hot-swap 2.5" NVMe drive bays 2 front hot-swap 2.5" SATA drive bays M.2: 1 M.2 PCIe 3.0 x4 NVMe slot (M-key)
Expansion Slots	Default 11 PCIe 5.0 x16 FHFL slots Option A 13 PCIe 5.0 x16 FHFL slots
On-Board Devices	SATA: SATA (6Gbps) Chipset: AMD SP5 Network Connectivity: 2 RJ45 10GbE
Input / Output	LAN: 1 RJ45 1 GbE Dedicated BMC LAN port 2 RJ45 10 GBASE-T LAN ports USB: 2 ports(rear) Video: 1 VGA port Serial: 1 COM port(rear)

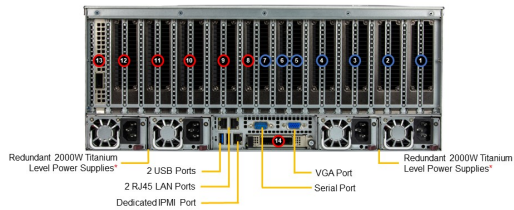
(Front View – System)



Front Panel

Drive Bay	Description
05 – 07	2 Hot-swap 2.5" SATA Drive Bays (default)
08 – 10	8 Hot-swap 2.5" NVMe Drive Bays (default)
11 – 24	24 Hot-swap 2.5" SATA/SAS Drive Bays (requires optional storage controller)

(Rear View – System)



Slot	Description
1 – 4	PCIe 5.0 x16
5 – 8	PCIe 5.0 x16
9	PCIe 5.0 x16 AOM (OCP 3.0 compatible)

*Full redundancy based on configuration and application load.

System Cooling	Fans: Up to 8 heavy duty fans with optimal fan speed control
Power Supply	4x 2000W Redundant (3 + 1) Titanium Level (96%) power supplies
System BIOS	BIOS Type: AMI 32MB SPI Flash EEPROM
Management	SuperCloud Composer®; Supermicro Server Manager (SSM); Supermicro Update Manager (SUM); Supermicro SuperDoctor® 5 (SD5); Super Diagnostics Offline (SDO); Supermicro Thin-Agent Service (TAS); SuperServer Automation Assistant (SAA) New!
PC Health Monitoring	CPU: 7 +1 Phase-switching voltage regulator Monitors for CPU Cores, Chipset Voltages, Memory FAN: Fans with tachometer monitoring Pulse Width Modulated (PWM) fan connectors Status monitor for speed control Temperature: Monitoring for CPU and chassis environment Thermal Control for fan connectors
Dimensions and Weight	Weight: Gross Weight: 100 lbs (45.3 kg) Net Weight: 65.5 lbs (29.7 kg) Available Color: Black Front & Silver Body
Operating Environment	Operating Temperature: 10°C to 35°C (50°F to 95°F) Non-operating Temperature: -40°C to 60°C (-40°F to 140°F) Operating Relative Humidity: 8% to 90% (non-condensing) Non-operating Relative Humidity: 5% to 95% (non-condensing)
Motherboard	Super H13DSG-O-CPU
Chassis	CSE-418G2TS